

ZETECH UNIVERSITY

UNIT NAME: PROGRAMMING WITH.NET Using C#.

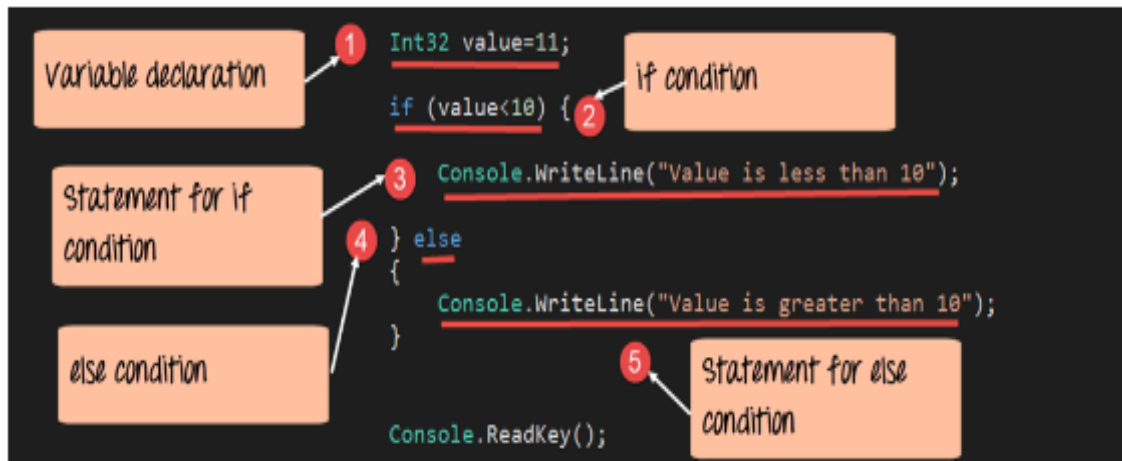
UNIT CODE: BIT 321

Lesson 3 Decision Making

- Decision Making in programming is similar to decision making in real life. In programming too, **a certain block of code needs to be executed when some condition.**

1) If statement

The if statement is used to evaluate a boolean expression before executing a set of statements. If an expression evaluates to true, then it will run one set of statements else it will run another set of statements.



```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace DemoApplication
{
    class Program
    {
        static void Main(string[] args)
        {
            Int32 value = 11;

            if(value<10)
            {
                Console.WriteLine("Value is less than 10");
            }
            else
            {
                Console.WriteLine("Value is greater than 10");
            }
            Console.ReadKey();
        }
    }
}

```

Problem Description

This C# Program checks if a given integer is Odd or Even.

Problem Solution

Here if a given number is divisible by 2 with the remainder 0 then the number is an Even number. If the number is not divisible by 2 then that number will be an Odd number.

```

/*
 * C# Program to Check whether the Entered Number is Even or Odd
 */
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace check1
{
    class Program
    {
        static void Main(string[] args)
        {
            int i;
            Console.Write("Enter a Number : ");
            i = int.Parse(Console.ReadLine());
            if (i % 2 == 0)
            {
                Console.WriteLine("Entered Number is an Even Number");
                Console.ReadLine();
            }
            else
            {
                Console.WriteLine("Entered Number is an Odd Number");
                Console.ReadLine();
            }
        }
    }
}

```

Program Explanation

In this C# program, we are reading the number using 'i' integer variable. If condition statement is used to check the number is even and odd. For even number the modulus of the value of 'i' variable by 2 is equal to zero, if the condition is true then print the statement as even number.

Parse(String) Method is used to convert the string representation of a number to its 32-bit signed integer equivalent.

C# if...else if (if-then-else if) Statement

- When we have only one condition to test, if-then and if-then-else statement works fine. But what if we have a multiple condition to test and execute one of the many block of code.

For such case, we can use if. Else if statement in C#. The syntax for if...else if statement is:

The program will determine the value of your grade using the if else if statement.

Writing C# program if-else-if using window form Application to grade student marks

We will be using **Window Form Application in C#**. It is used to develop a graphical user interface

Let's begin the Exercise:

Open your Microsoft Visual Studio 2010.

Select **File** Then **New Project**.

Select **C#** then **Windows Form Application** and click OK.

We will need the following controls:

- 1 label -
- 1 textbox
- 1 button

- **label**

to display text in a set location on the page. Label controls can also be used to add descriptive text to a Form to provide the user with helpful information. The Label class is defined in the System. Windows.

textbox

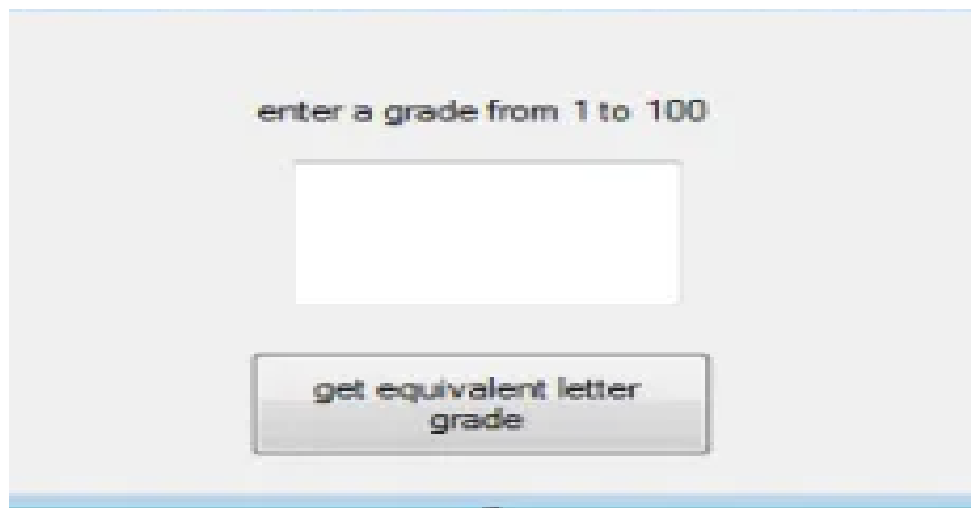
- **text**

A text box object is used to display text on a form or to get user input while a C# program is running.

- **button**

It allows the user to interact with the application or software. For example, if a user wants to exit from the current application so, he/she click the exit button which closes the application.

1. Open your Microsoft Visual Studio 2010.
2. Select File then New Project.
3. Select C# then Windows Form Application and click OK.
4. We will need the following controls:
 1. 1 label
 2. 1 textbox
 3. 1 button



Example form design

5. Double click the button. It will look like this.

```
private void button1_Click(object sender, EventArgs e)
{
}
}
```

6. Paste the code below between the { } open and close bracket

```
if (textBox1.Text == "")
{
    MessageBox.Show("enter a score from 1 to 100");
    textBox1.Focus();
}
else if (double.Parse(textBox1.Text) > 95 && double.Parse(textBox1.Text) <= 100)
{
    MessageBox.Show("equivalent is A");
}
else if (double.Parse(textBox1.Text) >= 90 && double.Parse(textBox1.Text) <= 94)
{
    MessageBox.Show("equivalent is A-");
}
else if (double.Parse(textBox1.Text) >= 87 && double.Parse(textBox1.Text) <= 89)
{
    MessageBox.Show("equivalent is B+");
}
else if (double.Parse(textBox1.Text) > 83 && double.Parse(textBox1.Text) <= 86)
{
    MessageBox.Show("equivalent is B-");
}
else if (double.Parse(textBox1.Text) >= 80 && double.Parse(textBox1.Text) <= 83)
{
    MessageBox.Show("equivalent is B");
}
else if (double.Parse(textBox1.Text) >= 75 && double.Parse(textBox1.Text) <= 79)
{
    MessageBox.Show("equivalent is C");
}
else if (double.Parse(textBox1.Text) < 75)
{
    MessageBox.Show("equivalent is F");
}
```

Code Explanation:

The user must enter a grade before clicking the button or else a message box will appear to notify the user that he or she must enter a grade from 1 to 100. If your grade is greater than 95 the equivalent is A, 90 to 94 is A-, 87 to 89 is B+, 84 to 86 is B-, 80 to 83 is B, 75 to 79 is C and less than 75 is F.

Note: you can also set your own range and value for this program.

7. Save and Test the project.



Student Grade Calculation using C#.net.....Video Link ****must watch to get concept in details (using windows form App)**

<https://www.youtube.com/watch?v=JzLGEgB3F7s>

C# Switch statement

Switch statement can be used to replace the if...else if statement in C#. The advantage of using switch over if...else if statement is the codes will look much cleaner and readable with switch.

- A **switch statement is significantly faster than an if-else ladder** if there are many nested if-else's involved.
- This is due to **the creation of a jump table for switch during compilation**. As a result, instead of checking which case is satisfied throughout execution, it just decides which case must be completed
- switch is to test **char** and **integer data type not (float, double, etc)** **if statements takes care of the other data types**.

The syntax of switch statement is:

```
switch (variable/expression)
{
    case value1:
```

```

        // Statements executed if expression(or variable) = value1

        break;

    case value2:

        // Statements executed if expression(or variable) = value1

        break;

    ... ..

    ... ..

    default:

        // Statements executed if no case matches

}

```

```
using System;
```

```
namespace Conditional
```

```

{
    class SwitchCase
    {
        public static void Main(string[] args)
        {
            char ch;

            Console.WriteLine("Enter an alphabet");

            ch = Convert.ToChar(Console.ReadLine());

            switch(Char.ToLower(ch))
            {
                case 'a':

```

```

        Console.WriteLine("Vowel");
        break;
    case 'e':
        Console.WriteLine("Vowel");
        break;
    case 'i':
        Console.WriteLine("Vowel");
        break;
    case 'o':
        Console.WriteLine("Vowel");
        break;
    case 'u':
        Console.WriteLine("Vowel");
        break;
    default:
        Console.WriteLine("Not a vowel");
        break;
    }
}
}
}
}

```

Output

Enter an alphabet

X

Not a vowel

EXAMPLE 2: Simple calculator program using C# switch Statement
using System;

namespace Conditional

```

{
    class SwitchCase
    {
        public static void Main(string[] args)
        {
            char op;
            double first, second, result;

            Console.Write("Enter first number: ");
            first = Convert.ToDouble(Console.ReadLine());
            Console.Write("Enter second number: ");
            second = Convert.ToDouble(Console.ReadLine());
            Console.Write("Enter operator (+, -, *, /): ");
            op = (char)Console.Read();

            switch(op)
            {
                case '+':
                    result = first + second;
                    Console.WriteLine("{0} + {1} = {2}", first, second, result);
                    break;

                case '-':
                    result = first - second;
                    Console.WriteLine("{0} - {1} = {2}", first, second, result);
                    break;

                case '*':
                    result = first * second;

```

```
        Console.WriteLine("{0} * {1} = {2}", first, second, result);
        break;

    case '/':
        result = first / second;
        Console.WriteLine("{0} / {1} = {2}", first, second, result);
        break;

    default:
        Console.WriteLine("Invalid Operator");
        break;

    }
}
}
```

OUTPUT

Enter first number: -13.11

Enter second number: 2.41

Enter operator (+, -, *, /): *

-13.11 * 2.41 = -31.5951

Revision Questions

1. Write a C# Sharp program to accept two integers and check whether they are equal or not.

Test Data :

Input 1st number: 5

Input 2nd number: 5

2. Write a C# Sharp program to check whether a given number is even or odd.

Test Data : 15

3. Write a C# Sharp program to find the eligibility of admission for a professional course based on the following criteria: Go to the editor

Marks in Maths ≥ 65

Marks in Phy ≥ 55

Marks in Chem ≥ 50

Total in all three subject ≥ 180

or

Total in Math and Subjects ≥ 140

Test Data :

Input the marks obtained in Physics :65

Input the marks obtained in Chemistry :51

Input the marks obtained in Mathematics :72